

NORMALIZATION

Normalization is a technique to remove or reduce redundancy from a table

StudentID	StdName	Age
01	Asad	20
02	Saleem	25
01	Asad	20

Example No.1

- Row level Duplication
- By setting Primary key ,duplication can be reduced

<https://www.youtube.com/watch?v=5GDTIUVIHB8>

EXAMPLE II

StdID	Sname	CourseID	CourseName	FacID	FacName	Salary
01	Asad	C1	DBMS	F1	Mr.A	30000
02	Saleem	C2	Java	F2	Mr.B	40000
03	Khalid	C1	DBMS	F1	Mr.A	30000
04	Rasheed	C1	DBMS	F1	Mr.A	30000

- Column Level Duplication

Issues :

- (i) Insertion Anomaly
- (ii) Deletion Anomaly
- (iii) Updation Anomaly

Solution :

Separate tables such as

- (i) Student
- (ii) Course
- (iii) Faculty

1ST NORMAL FORMAL

- As per EF codd ,table should not contains multivalued attributes

Roll No	Name	Course
1	Ahmed	C/C++
2	Saleem	Java
3	Asad	C/DBMS

SOLUTION

Roll No	Name	Course
1	Ahmed	C
1	Ahmed	C++
2	Saleem	Java
3	Asad	C
3	Asad	DBMS

1ST NORMAL FORMAL CONT...

SOLUTION 02

Student

Roll No	Name	Course1	Course 2
1	Ahmed	C	C++
2	Saleem	Java	Null
3	Asad	C	DBMS

SOLUTION 03

Student

Roll No	Name
1	Ahmed
2	Saleem
3	Asad

Course

Roll No	Course
1	C
1	C++
2	Java
3	C
3	DBMS

Second Normal Form 2NF

Any non key field should entirely depend on its primary key

1 - Should be in 1NF

2 - No PARTIAL DEPENDENCY

3 - Occurs when there is composite key

Second Normal Form 2NF

Result

StudentId	Course	Name	Marks	Teacher
201	Software Architecture	Saghir	85	A
202	Software Design	Harris	90	B
201	Quality Assurance	Saghir	75	G
204	Enlish Language	Andy	63	O
205	History	Simon	74	L
206	Project Managent	Sam	93	G
205	Software Architecture	Simon	70	A
208	Quality Assurance	Taylor	61	G

Primary
key

Primary
key

Non Key Columns

TEACHER
partially
depends on
COMPOSITE KEY

← **violating 2NF**

SOLUTION

Result

StudentId	Course	Name	Marks
201	Software Architecture	Saghir	85
202	Software Design	Harris	90
201	Quality Assurance	Saghir	75
204	Enlish Language	Andy	63
205	History	Simon	74
206	Project Managent	Sam	93
205	Software Architecture	Simon	70
208	Quality Assurance	Taylor	61

Teacher

Course	Teacher
Software Architecture	A
Software Design	B
Quality Assurance	G
Enlish Language	O
History	L
Project Managent	G

Third Normal Form 3NF

Any non key field dependent on other non key field

- 1 - Should be in 1NF, 2NF
- 2 - No TRANSITIVE DEPENDENCY
- 3 - Occurs when you can guess value of any column from non key column

Third Normal Form 3NF

Exam

StudentId	Name	ExamType	MaxMarks
201	Saghir	Viva	20
202	Harris	Theroy	100
203	Maxwell	Practical	50
204	Andy	Practical	50
205	Simon	Viva	20
206	Sam	Theroy	100
207	Jim	Theroy	100
208	Taylor	Practical	50

Primary
key

violating 3NF

MaxMarks
transitively
depends on
Examtype

we can Guess value of MaxMarks from ExamType

VIVA = 20

THERORY=100

PRACTICAL=50

Exam

StudentId	Name	ExamType	MaxMarks
201	Saghir	Viva	20
202	Harris	Theroy	100
203	Maxwell	Practical	50
204	Andy	Practical	50
205	Simon	Viva	20
206	Sam	Theroy	100
207	Jim	Theroy	100
208	Taylor	Practical	50

209 NEW Practical 50

SOLUTION

Exam

StudentId	Name	ExamType
201	Saghir	Viva
202	Harris	Theroy
203	Maxwell	Practical
204	Andy	Practical
205	Simon	Viva
206	Sam	Theroy
207	Jim	Theroy
208	Taylor	Practical

Marks

ExamType	MaxMarks
Viva	20
Theroy	100
Practical	50

Link both tables by ExamType common table

3rd NF EXAMPLE NO.02

Match

Match#	Teams	Ground	Capicity
01	Aus v Eng	MCG	80000
02	Zim v Eng	Hobart	30000
03	Sl v Ind	SCG	65000
04	Pak v Wi	MCG	80000
05	SA v NZ	Hobart	30000
06	Ausv Ind	Perth	45000
07	Pak v Ind	Canbera	10000
08	Eng v Sl	MCG	80000

Capicity
transitively
depends on
Ground

violating 3NF

SOLUTION

Match

Match#	Teams	Ground
01	Aus v Eng	MCG
02	Zim v Eng	Hobart
03	Sl v Ind	SCG
04	Pak v Wi	MCG
05	SA v NZ	Hobart
06	Ausv Ind	Perth
07	Pak v Ind	Canbera
08	Eng v Sl	MCG

Capicity

Ground	Capicity
MCG	80000
Hobart	30000
SCG	65000
Perth	45000
Cambera	10000

3rd NF EXAMPLE NO.03

Employee

EmployeeId	Name	Designation	Salary
201	Saghir	Manager	80000
202	Harris	Lecturer	40000
203	Maxwell	Manager	80000
204	Andy	Lecturer	40000
205	Simon	Worker	10000
206	Sam	Worker	10000
207	Jim	Lecturer	40000
208	Taylor	Lab Assistant	25000

Salary
transitively
depends on
Designation

Whenever you can guess the value from non key column its violating 3nf

SOLUTION

Employee

EmployeeId	Name	Designation
201	Saghir	Manager
202	Harris	Lecturer
203	Maxwell	Manager
204	Andy	Lecturer
205	Simon	Worker
206	Sam	Worker
207	Jim	Lecturer
208	Taylor	Lab Assistant

Salary

Designation	Salary
Manager	80000
Lecturer	40000
Worker	10000
Lab Assistant	25000

3rd NF EXAMPLE NO.04

Matches

Match#	Teams	Tickets	PerTicket	Earning
01	Aus v Eng	1000	200	200000
02	Zim v Eng	1500	200	300000
03	Sl v Ind	2000	200	400000
04	Pak v Wi	1000	200	200000
05	SA v NZ	1500	200	300000
06	Aus v Ind	2000	200	400000
07	Pak v Ind	1000	200	200000
08	Eng v Sl	1200	200	240000

Earning is depending on number of tickets sold

1000 * 200 = 200000 its calculations,violating 3NF